

Affan Ahmed Khan

Full-Stack Developer | Real-Time Systems Engineer

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PROFESSIONAL SUMMARY

Full-Stack Developer with hands-on experience shipping real-time, production-grade web applications using React, Node.js, and WebSockets. Focused on backend architecture, event-driven systems, and performance optimization — with a track record of building scalable SaaS platforms from the ground up.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C++, C#

Frontend: React, TypeScript, Tailwind CSS, Framer Motion, Zustand, TanStack Query

Backend: Node.js, Express, Prisma, PostgreSQL, REST APIs, Socket.io

Systems: Redis, BullMQ, AWS S3, Firebase, rrweb, Chrome Extension APIs

Data: SQL, Pandas, NumPy, Matplotlib, Seaborn

Architecture: Real-Time Systems, RBAC, JWT Auth, Event-Driven Design

EXPERIENCE

Federal Board of Revenue (FBR)

Jul 2025 – Sep 2025

Data & Compliance Intern

- Validated and cleaned taxation datasets spanning 10,000+ records, reducing reporting discrepancies by an estimated 30%.
- Built Python scripts to automate repetitive data-checking workflows, cutting manual processing time by several hours per week.
- Produced structured compliance reports used directly in internal audit processes across multiple departments.

PROJECTS

NeoPlane — Real-Time Communication Platform

[Live Demo](#)

React, Node.js, Socket.io, Prisma, PostgreSQL, AWS S3

- Engineered a low-latency messaging system using Socket.io, supporting real-time broadcast across concurrent rooms with sub-100ms delivery.
- Designed a secure backend (Express + Prisma + JWT) with optimized relational queries, cutting average response time by ~40% via query batching and indexing.
- Built a media pipeline with AWS S3 + Sharp achieving ~60% file size reduction through automatic image compression before storage.
- Eliminated ~35% of redundant API calls using Zustand + TanStack Query for shared client-side state caching.

Nexus — Enterprise Task Management Platform

[Live Demo](#)

React, TypeScript, Node.js, PostgreSQL, Prisma, Socket.io

- Built a real-time Kanban system with live presence tracking and activity streams, handling concurrent multi-user updates without conflict.
- Implemented full RBAC with JWT + refresh token rotation, securing role-scoped access across admin, member, and guest tiers.
- Applied optimistic UI updates via React Query + Zustand, reducing visible loading states by ~50% and improving perceived responsiveness.
- Designed analytics dashboards with paginated audit logs for efficient querying over large event histories.

E-Commerce Engine & Analytics Platform

[Live Demo](#)

TypeScript, Node.js, PostgreSQL, Redis, BullMQ

- Architected a high-throughput REST API handling complex relational product, order, and inventory data with normalized schema design.
- Implemented async job pipelines with Redis + BullMQ for order processing and notification workflows, decoupling heavy tasks from the request cycle.
- Designed a KPI analytics engine using snapshot strategies and optimized aggregation queries, delivering dashboard data in under 200ms.

Universal Web Debugger — Chrome Extension

TypeScript, Chrome APIs, Shadow DOM, rrweb

- Built a browser extension performing runtime audits across performance, accessibility, and CSS conflict categories for any live webpage.
- Used Shadow DOM for full UI isolation, ensuring zero style leakage between the extension panel and the host page.
- Integrated rrweb session replay to capture and replay user interaction flows, reducing debugging time for complex UI issues.

EDUCATION

NUST — Military College of Signals (MCS), Islamabad

Expected 2027

BS Software Engineering CGPA: 3.07

- Relevant Coursework:** Machine Learning, Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Systems